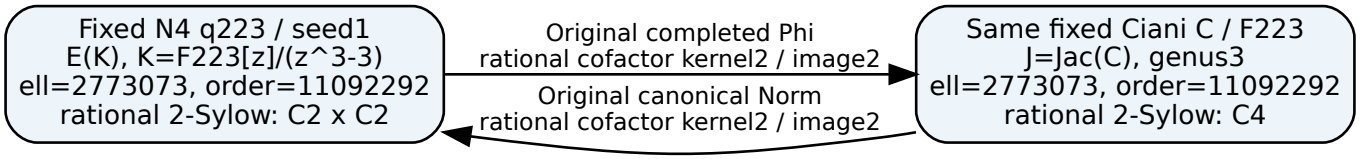
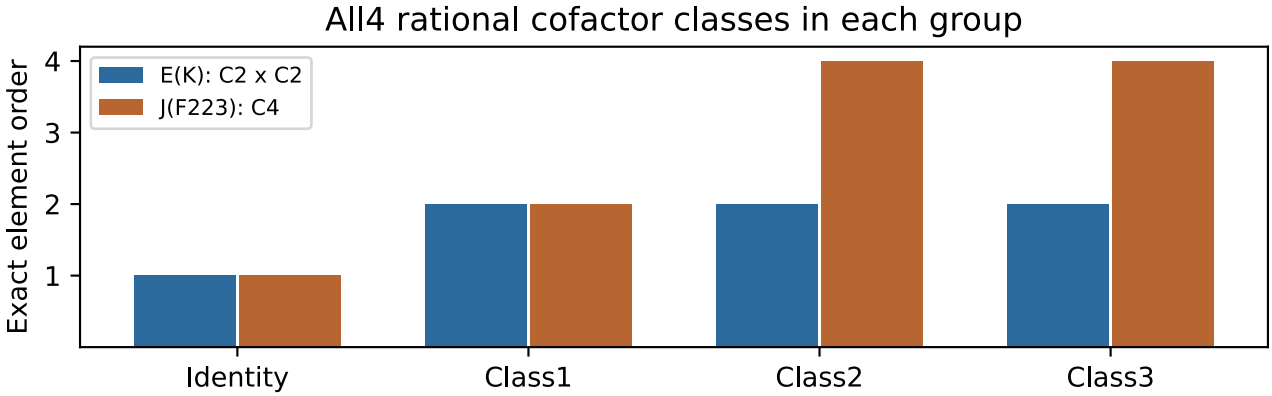
The actual rational two-primary groups

2026-11-13 · fixed N4 q223 / seed1 · unsigned / partial formalization





Fixed native observation510d03, unsigned: original Picard arithmetic closes a four-element group from all224 rational C point classes projected byell and the actual Phi(E(K)[2]) images. Exact orders are1,2,4,4, all16 sums,64 associative triples and inverses pass. The written global Ciani isogeny supplies the order4ell bound, so this is the entire rational2-Sylow,C4. Native annihilators alone would not establish that conclusion.

The independent source engine finds all three distinct K-rational roots of the elliptic cubic, hence E(K)[2]=C2×C2. Actual Phi and Norm each have rational cofactor kernel2/image2, all16 homomorphism pairs and both[2] compositions pass. The nonzero transported odd generator separately has exact prime orderell. This difference is consistent with a degree-even isogeny.

Native02 passes one unfiltered serial test, all2318 selected input hashes unchanged. All four native JSON records match01 exactly as JSON. The first capture omitted direct V2 dependency snapshots; whole01 and correction1311d9 remain preserved. Its incomplete provenance was not promoted to full closure. This is known-case correctness development with a reused cache, not a performance experiment.

Why the full two-torsion needs degree six

Written derivation: the Ciani quotient maps give an F223-isogeny J~Res(K/F223)E. With q=223,t=q³+1−11092292=−2724, the Weil polynomial is P(T)=T⁶+2724T³+11089567. Its integral2-adic characteristic polynomial reduces to T⁶+1 on the actual six-dimensional geometric J[2]. This use of isogeny invariance does not transfer two-torsion through a degree-even isogeny. Cayley-Hamilton gives F⁶=1.

There are64 geometric2-torsion points and exactly2 fixed by F, because the rational2-Sylow is C4. If F³=1, all remaining orbits have length3, requiring64=2+3c, impossible. If F²=1, the minimal polynomial divides(T+1)²; the characteristic polynomial must be(T+1)⁶=T⁶+T⁴+T²+1, different from T⁶+1. Thus F has exact order6: the full geometric2-torsion has minimal field F223⁶, not the original cubic field.

Partial Lean proof map: four faithful declarations cover exact trace/parity, the integer orbit obstruction, the annihilating-polynomial implication on an arbitrary exponent-two module, and the divisor-of-six conclusion. Both unchanged original producer4.35 and consumer4.34.1 gates pass4/0trusted. Exact original PCC local bindings4e7bfe/cc7273/abcfec/5facfe pass,unsigned. Foundations are propext/Quot.sound only; no proof holes, custom axioms or native_decide. No fresh lean4checker result is claimed.

The full geometric bridges remain written: actual Jacobian/torsion definitions, Ciani isogeny and Weil polynomial, Tate module reduction, Cayley-Hamilton, Frobenius orbit decomposition, characteristic/minimal polynomial and minimal splitting field. The module lemma assumes the faithful polynomial identity, not the desired sixth-iterate conclusion. Integer lemmas do not discharge these geometric obligations.

Decision: do not assume geometric J[2] is cubic-rational or that Frobenius is semisimple of order3. A rational isogeny can still have a Frobenius-stable kernel with non-rational individual points. Actual geometric basis/ action, stable maximal-isotropic kernel census, a second nondual map and target equation remain open. The next discriminating construction is explicit geometric2-torsion over F223⁶ or a coordinate kernel-descent construction. No computational advantage is established.

Reproducible evidence

Native02/510d03; all224 point projections and all4 classes in adjacent compressed native JSON. Full equations: V2_MAPS_01.md / unit-acceptance-01/instance.json. Written proof and exact formal gaps: two-primary/DERIVATION-01.md. Four local PCC bindings: evidence/two-primary-bindings-01/claims.json. No new nondual map or cost comparison.